

Mahmoud Amer Fawzy

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EDUCATION

Bachelor of Engineering in Electronics and Communications (ECE)
Aswan University

2019 – 2024
Aswan

SUMMARY

Electronics and Communications Engineering graduate with knowledge of embedded systems, networking, IT, and software development. Proficient in C, C++, Python, Windows Server Administration, and Cisco networking, with hands-on project experience and strong problem-solving skills. Eager to apply technical knowledge and grow in a dynamic engineering environment.

SKILLS

Embedded Systems

- I have a strong background in embedded systems, including **Embedded C programming**, **microcontroller fundamentals**, and digital electronics gained through my ECE studies. I understand hardware and software interaction and have developed problem-solving skills in embedded system design through academic projects.

IT Technical Skills

- Windows administration**, computer hardware, Windows installation and configuration, hardware and software troubleshooting, system maintenance, network connectivity, virtualization, basic security and cloud concepts, and **IT ticketing systems (ServiceNow)**. Hands-on experience providing technical support, configuring systems, and resolving end-user issues.

Programming

- Proficient in **C**, **C++**, and **Python**, with experience in scripting, problem-solving, and basic software development concepts.

Networking

- TCP/IP, OSI Model, IPv4, **Subnetting**, **DHCP**, **DNS**, **VLANs**, NAT/PAT, ACLs, WLAN, and Cisco Packet Tracer. Hands-on experience configuring and troubleshooting network connectivity, routing and switching fundamentals, and resolving common network issues. Familiar with network design principles and basic network security concepts.

COURSES

CCNA

Coursera

Completed CCNA coursework covering TCP/IP, IPv4 addressing, subnetting, Routing & Switching, VLANs, DHCP, DNS, NAT/PAT, ACLs, WLANs, and network troubleshooting. Gained hands-on experience configuring and troubleshooting networks using Cisco Packet Tracer.

Windows Server Administration Course

Completed training in Windows Server installation and configuration, Active Directory Domain Services (AD DS), user and group management, Organizational Units (OU), Group Policy (GPO), DNS, DHCP, domain management, NTFS and shared folder permissions, Remote Desktop Services, Hyper-V virtualization, and WSUS. Gained hands-on experience administering Windows Server environments in a virtualized lab.

CompTIA A+

Coursera

Completed CompTIA A+ coursework covering PC hardware, Windows installation and administration, hardware and software troubleshooting, networking fundamentals (TCP/IP, DNS, DHCP), mobile devices, security fundamentals, virtualization, cloud concepts, and IT operational procedures. Gained hands-on experience in system configuration, technical support, and troubleshooting end-user issues.

Data Analysis

NTI

- Learned Python for data analysis, worked with Pandas and NumPy for data manipulation, cleaned and sorted Excel datasets, and created visualizations using Matplotlib.

Embedded Systems Diploma

ITI

Completed Training C Programming, Microcontrollers, AVR Interfacing, Real-Time Operating Systems (RTOS), Embedded Systems Concepts, Hardware/Software Interfacing, and Embedded Development Tools.

Microsoft 365 (MS-900)

Microsoft 365 Administration, Microsoft Entra ID, Intune, Exchange Online, Teams, SharePoint, OneDrive, Microsoft Defender, Licensing, Security & Compliance

PROJECTS

Smart Robotic Wheelchair (Graduation Project – Team Based)

- I worked on a Smart Robotic Wheelchair as a graduation team project, where I contributed to developing a multi-mode control system. My role focused on interfacing healthcare sensors and collecting real-time data using a microcontroller. I also implemented data transmission to a mobile application using a real-time database, enabling IoT-based communication features such as mode selection and speed control through the app.

Smart Home Simulation System

- I also developed a Smart Home Simulation System using embedded system concepts. The system included a keypad-based password authentication system with an LCD for door control, an automatic lighting system based on light sensors, and an automatic AC control system based on temperature sensor readings.

Automatic Irrigation System

- Additionally, I designed an Automatic Irrigation System that controls plant watering based on soil moisture levels. The system automatically activates and deactivates a water valve depending on sensor input, simulating real-time environmental monitoring and automated response behavior.

IT Support Projects

- I have completed basic CCNA networking labs and simulations, gaining practical understanding of IP addressing, subnetting, and basic routing and switching concepts through Cisco Packet Tracer.

Small Office Network Design and Simulation

- I also developed a **Small Office Network Design and Simulation** using **Cisco Packet Tracer**. The project involved designing a small office network with VLAN segmentation, configuring trunk ports and inter-VLAN routing, deploying network servers, assigning IP addressing, and validating end-to-end connectivity through network testing and troubleshooting.